MILITARY SPECIFICATION SHEET

RELAYS, HYBRID, SENSORS, ESTABLISHED RELIABILITY, TYPE III (AC UNDERVOLTAGE SENSOR), CLASS B, 2PDT, LOW LEVEL TO 10 AMPERES, HERMETICALLY SEALED (POTTED)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the relays described herein shall consist of this specification and the latest issue of MIL-R-28894.

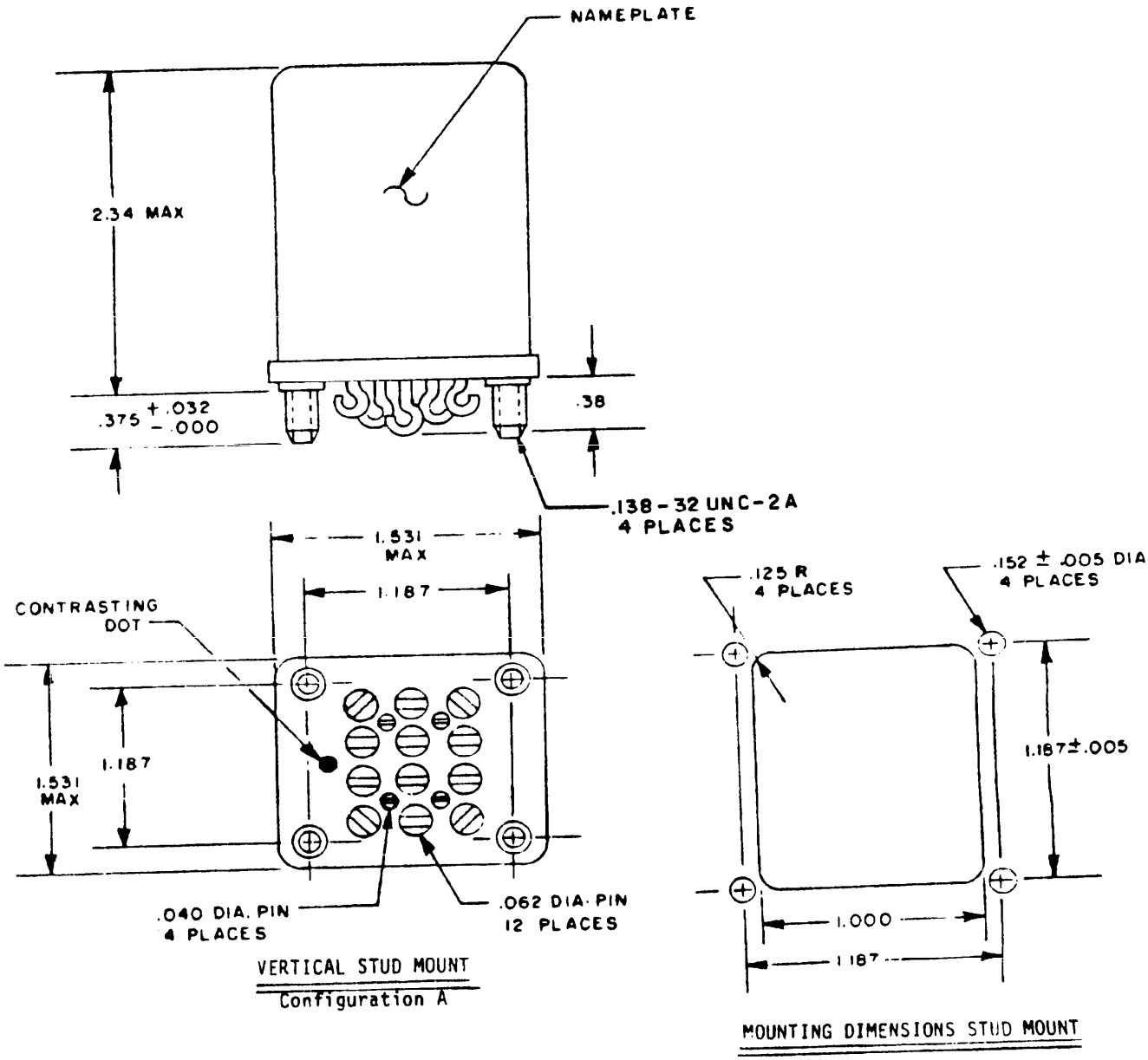


FIGURE 1. Dimensions and configuration.

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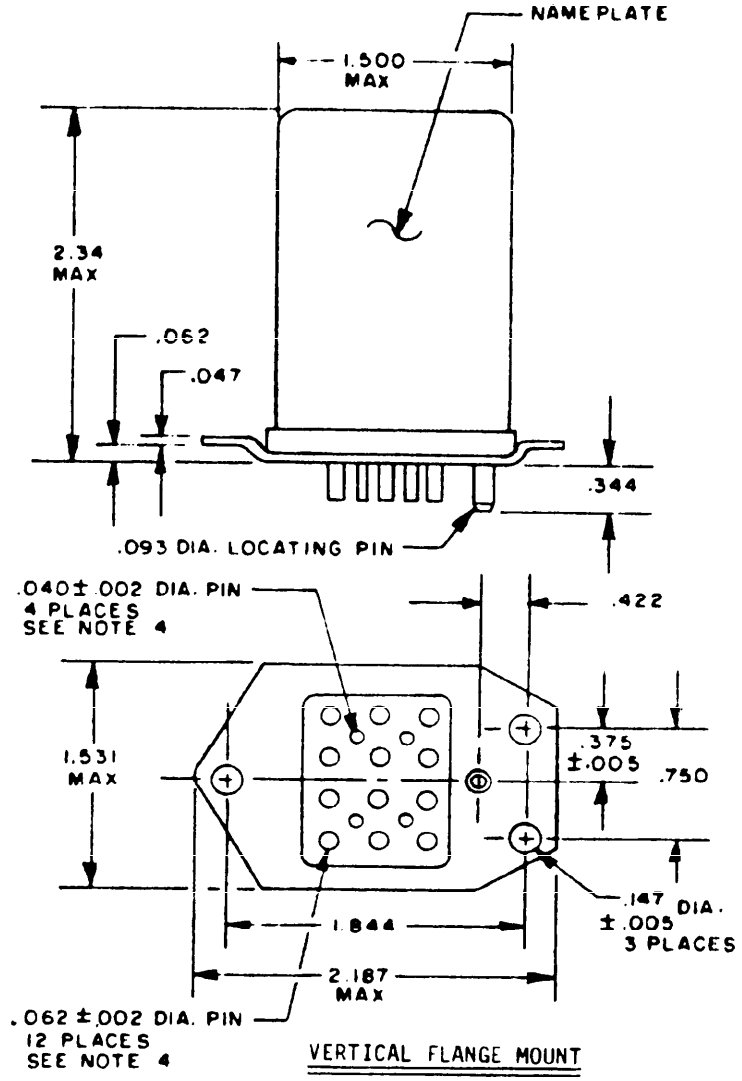


FIGURE 1. Dimensions and configuration - Continued.

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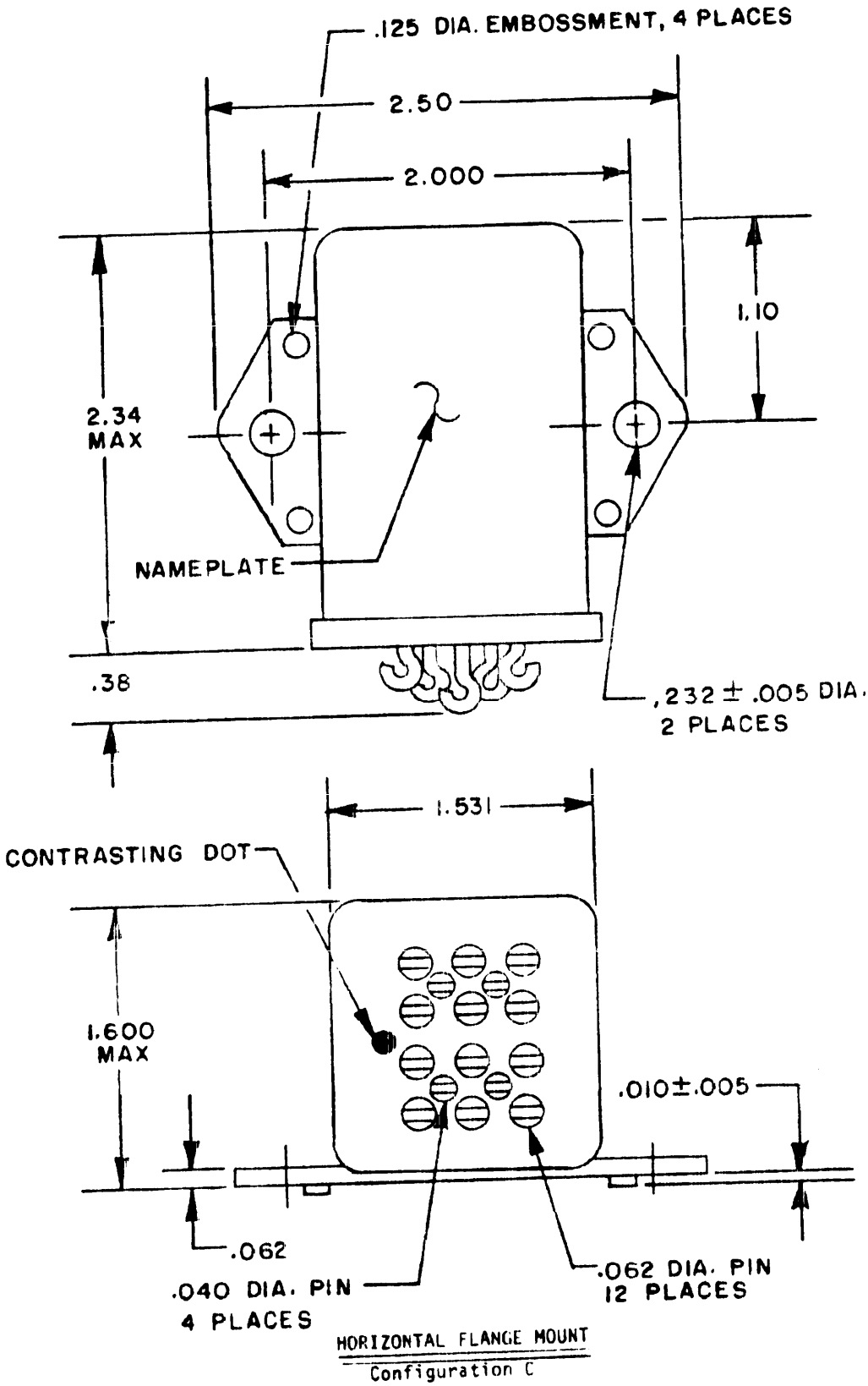
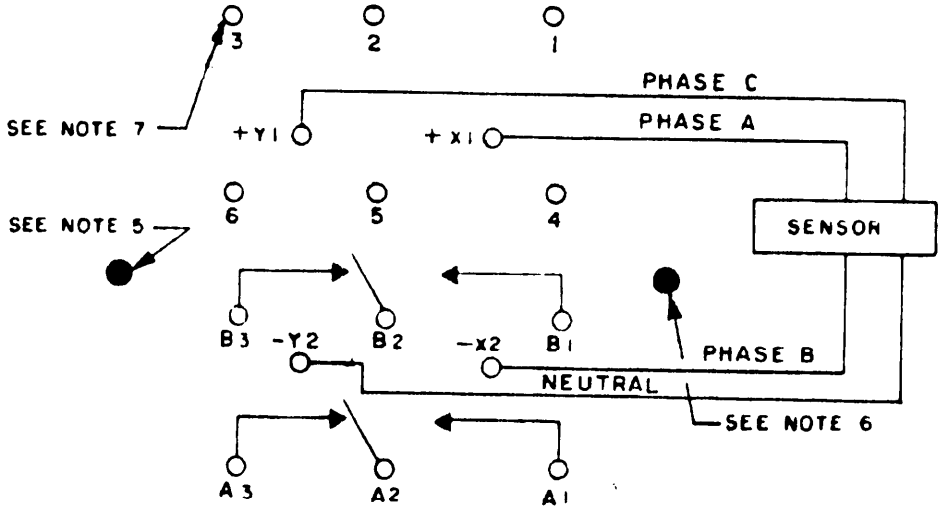


FIGURE 1. Dimensions and configuration - Continued.

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CIRCUIT DIAGRAM TERMINAL VIEW

Pin	Input Connection
+X1	Phase A
-X2	Phase B
+Y1	Phase C
-Y2	Neutral

Inches	mm	Inches	mm
.000	0.00	.375	9.53
.002	0.05	.38	9.7
.005	0.13	.750	19.05
.010	0.25	.922	23.42
.032	0.81	1.000	25.40
.040	1.02	1.10	27.9
.062	1.58	1.187	30.15
.093	2.36	1.500	38.10
.125	3.18	1.531	38.89
.138	3.51	1.844	46.83
.147	3.73	1.887	47.93
.152	3.86	2.187	55.55
.218	5.54	2.34	59.4
.344	8.74		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Terminal numbers shown above for reference only. Numbers do not appear on header.
4. Finish shall be gold plate per MIL-G-45204, type II, class 1, underplating shall be nickel 50 to 150 microinches.
5. A contrasting dot shall appear on header and circuiting diagram for vertical stud mount and horizontal flange mount only.
6. A locating pin designator shall appear on circuit diagram for vertical flange mount only.
7. Terminals 1, 2, 3, 4, 5, and 6 are non-functional leads.
8. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±.03 (0.8 mm) for two place decimals.

FIGURE 1. Dimensions and configuration - Continued.

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REQUIREMENTS:

Input data:

Input voltage range: 90 V ac to 124 V ac, line to neutral.

Input frequency range: 400 Hz ±20 Hz, 3 phase wye.

Sensing accuracy range (based on a true sinusoidal input waveform): ±2% maximum.

Operating mode: The output relay energizes when the voltages of all three phases are above the trip point and de-energizes when the voltage of any phase drops below the trip point.

Operating data:

Pickup time (electromechanical relay): 50 milliseconds maximum when sensed voltage is at or above the trip point.

Dropout time (electromechanical relay): 50 milliseconds maximum when sensed voltage is below the trip point.

Current: Maximum operating current of 75 milliamperes.

Contact data:

Configuration: 2PDT, 10 amperes, integral class B relay.

Load ratings:

High level (relay case grounded):

Resistive:
10 amperes at 28 V dc, 115 V ac 400 Hz and 115/200 V ac 400 Hz.
2.5 amperes at 115 V ac 50/60 Hz and 115/200 V ac 50/60 Hz. 1/

Inductive:
8 amperes at 28 V dc, 115 V ac 400 Hz and 115/200 V ac 400 Hz (20,000 cycles minimum life).
2.5 amperes at 115 V ac 50/60 Hz and 115/200 V ac 50/60 Hz. 1/

Motor:
4 amperes at 28 V dc, 115 V ac 400 Hz and 115/200 V ac 400 Hz.
2 amperes at 115 V ac 50/60 Hz and 115/200 V ac 50/60 Hz. 1/

Lamp:
2 amperes at 28 V dc, 115 V ac 400 Hz, and 115/200 V ac 400 Hz.
1 ampere at 115 V ac 50/60 Hz and 115V/200 V ac 50/60 Hz. 1/

Low level: 10 to 50 microamperes at 10 to 50 millivolts (dc or peak ac).

Intermediate current: Applicable.

Contact voltage drop:

High level:

Before life: 0.150 volt maximum.

After life: 0.175 volt maximum.

Low level:

Before life: 0.05 ohm maximum.

After life: 0.15 ohm maximum.

1/ For 50/60 Hz, overload is not applicable and life shall be 10,000 cycles.

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Contact bounce: 1.0 millisecond maximum (applicable to failure rate level "L").

Contact stabilization time: 1.5 milliseconds maximum (applicable to failure rate levels "M", "P", and "R").

Overload: 40 amperes dc, 60 amperes ac.

Electrical data:

Insulation resistance: 100 megohms minimum initial; 50 megohms after environmental or life tests.

Dielectric withstanding voltage:

	Initial	After life tests
Input to case:	1000 V rms (60 Hz)	1000 V rms (60 Hz)
All other points:	1250 V rms (60 Hz)	1000 V rms (60 Hz)

	Altitude	
	80,000 ft	300,000 ft
Input to case:	350 V rms (60 Hz)	500 V rms (60 Hz)
All other points:	350 V rms (60 Hz)	500 V rms (60 Hz)

Voltage trip points sensing range: 1 V dc to 50 V dc.

Voltage transients: MIL-STD-704, CAT-B.

Timing cycle: Not applicable.

Environmental data:

Temperature range: -55°C to +125°C.

Thermal shock: Applicable.

Vibration: MIL-STD-202, method 204, test condition D, except frequency shall be 5 to 2,000 Hz.

Vibration, random: MIL-STD-202, method 214, test condition IG, 15 minutes each plane.

Shock: MIL-STD-202, method 213, test condition A (50G's).

Resistance to soldering heat: Applicable.

Acceleration: Applicable.

Physical data:

Seal: Hermetic, potted.

Dimensions and configuration: See figure 1.

Terminal strength: Method 211 of MIL-STD-202.

Pull test: Test condition A, 10 pound pull.

Twist test: Test condition D.

Solderability: Applicable.

Weight: 284 grams (10 ounces) maximum.

Identification marking (full): Applicable.

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Life test requirements:

High level: 100,000 cycles minimum.

Low level: 100,000 cycles minimum plus 400,000 mechanical life.

Qualification inspection:

Qualification inspection and sample size: See table I.

TABLE I. Qualification inspection and sample size. 1/

Single submission	Group submission	
18 units plus 1 unsealed unit for level L at C = 0 2/ 34 units plus 1 unsealed unit for level M at C = 0 2/ Qualification inspection as applicable.	M28894/10-B090 M28894/10-B124	18 units plus 1 unsealed unit for level L at C = 0 2/ 34 units plus 1 unsealed unit for level M at C = 0 2/ Qualification inspection as applicable.
	M28894/10-A100 M28894/10-A105 M28894/10-C110 M28894/10-C115	2 units each part number Qualification inspection, group 1.

1/ For retention of qualification or extension of qualification to lower failure rate levels, all life test data accumulated on MIL-R-28894/11, MIL-R-28894/12, MIL-R-28894/13 maybe used in addition to MIL-R-28894/10 data.

2/ The number of units required for qualification testing will be increased as required in group V of the qualification inspection table of MIL-R-28894 if the contractor elects to test the number of units permitting one or more failures.

Qualification inspection (reduced testing) and sample size: See table II. If the relays produced for MIL-R-28894/10 are similar in construction and design, except for the number of output contacts and type of voltage sensed, to the relays produced for MIL-R-28894/11, MIL-R-28894/12 and MIL-R-28894/13, then reduced testing for qualification of MIL-R-28894/10 relays may be performed concurrent with or subsequent to successful qualification of MIL-R-28894/11, MIL-R-28894/12 and MIL-R-28894/13.

TABLE II. Qualification inspection (reduced testing).

Inspection
2 units each dash number specified in table I Group I of qualification inspection table plus shock and vibration 1 unsealed sample unit for internal examination

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Part number: Consists of the basic number of this specification sheet and a dash number denoting mounting configuration, voltage trip point, and failure rate level.

Example: M28894/10- A 110 L

- Basic specification number.
- Mounting configuration A, B, or C (see figure 1).
- Voltage trip point - Values may range from 090 to 124 V ac. Specify trip point using three digits. Example, for a trip point of 110 V dc, specify 110.
- Failure rate level (see table III).

TABLE III. Codes for failure rate level.

Code letter	Failure rate level (% per 10,000 cycles)
L	3.0
M	1.0
P	0.1
R	0.01

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AV
Navy - OS, SH

Preparing activity:
Navy - EC

Agent:
DLA - ES

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